

Create the 4 servers:

1st server: Install the java, git and Jenkins repo and Jenkins and access the Jenkins site, and install the docker in it and run the sonarqube. And access the sonarqube site with port 9000 with Jenkins server ip.

Command: `docker run -d --name sonar -p 9000:9000 sonarqube:lts-community`

Now Plugins in the Jenkins are pipeline stage view, deploy to container, nexus artifact uploader, sonar scanner, docker pipeline

And set up the sonarqube project, select "maven" in the "create Jenkins file option" and create the "global analysis token"

After generating the key, copy the key and then add it the "manage Jenkins → settings → sonarqube", add the credential with secret text. And also add sonarqube "mysonar" and maven "mymaven" in the tools section in the manage Jenkins. Because these are declared in the pipeline syntax.

2nd server: Install java, git, docker, trivy, docker compose (use script), swarm, nexus, in the opt folder and access the nexus site with the "docker manager" server with port 8081, and give the permission "chmod 777 /var/run/docker.sock" add this server in the manage Jenkins in the node section because we have to run the pipeline there. Give private ip and use ssh connection and give the pem key for the authentication.

Generate the pipeline script for artifact stage and image push stage (here give original docker cred)

Also install the docker compose, swarm in the server and then join other two servers with token. "docker swarm init"

we must use "devopsdb" as a service name for the database because developer integrated like this only, then only app and database will communicate.

Now application got accessed and create account. and to check the user details in the db cont but using these commands. Password from docker file of db.

`mysql -u root -p`, show databases; use accounts; show tables; select * from user; use all these commands inside the container.

In this project

Code, cqa, build, artifact (nexus), image creation, image scan, image push, deploy.

Here the application is a java application, which has two services (fe+be) and db. Here we use the docker to host the application.

Architecture: 4 servers one for jenkins (controller), second for docker manager (it acts as a jenkins agent as well where actual pipeline is executed), and other two servers are docker nodes, the three servers become the docker cluster.

Here tools used is

Git, GitHub, Jenkins, Sonarqube, maven, nexus, trivy, Docker Stack, cloud watch

aws | service Actions | Ask Amazon Q | United States (N. Virginia) | Melagiri Jayakrishna (529921977491) | Melagiri Jayakrishna

EC2 > Instances

Instances (4) info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Filter Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
☐	JENKINS (CONTROLLER)	i-0b9a83120ca84be10	Running	m7i-flex.large	3/3 checks passed	us-east-1d	ec2-3
☐	DOCKER-MANAGER (AGENT)	i-047ba74b946cc01ba	Running	m7i-flex.large	3/3 checks passed	us-east-1d	ec2-5
☐	NODE-1	i-025659ed3ca4035d5	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-5
☐	NODE-2	i-0c0eeea5da14e00dd	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-5

All instances

Instance alarm states

In alarm

None Show instances in alarm

Service health

EC2 service status

Operating normally

Health events affecting EC2 instances

Open events

None Show instances

Upcoming events

None Show instances

CloudShell | Agent Toolkit for AWS | Feedback | Console Mobile App | © 2026, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences

← → ↻ | Not secure | 35.175.238.49:9000/dashboard?id=javaproject | Search for projects... | A

sonarqube | Projects | Issues | Rules | Quality Profiles | Quality Gates | Administration

Devops Webapp | main | Last analysis of this Branch had 1 warning | October 9, 2026 at 9:33 PM | Version v2

Overview | Issues | Security Hotspots | Measures | Code | Activity | Project Settings | Project Information

QUALITY GATE STATUS

Passed

All conditions passed.

MEASURES

New Code	Overall Code
34	Reliability
2	Security
13	Security Review
2d 1h	Maintainability
6.7%	Unit Tests
4.8%	Duplicated Blocks

← → ↻ | Not secure | 52.91.65.255:8081/#browse/browse:nexusrepo | Search components or CVEs... | A

sonatype nexus repository | Community Edition

Browse | nexusrepo

Upload component | HTML View | Advanced search...

com

visualpathit

vprofile

v2

vprofile-v2.war

vprofile-v2.war.md5

vprofile-v2.war.sha1

The screenshot shows the Docker Hub interface for the repository 'imjk04/pr2'. The page is divided into a left sidebar with navigation links (Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, Storage) and a main content area. The main content area includes a header with the repository name and a 'General' tab. Below the header, there is a section for 'Tags' with a table listing two tags: 'app' and 'db'. The table columns are Tag, OS, Type, Pulled, and Pushed. The 'app' tag is the latest, pushed 1 minute ago. To the right of the tags section, there is a 'Docker commands' section showing the command 'docker push imjk04/pr2:tagname'. At the bottom of the page, there is a 'buildcloud' banner with the text 'Build with Docker Build Cloud' and a description of its features.

hub.docker.com/repository/docker/imjk04/pr2/general

New Get \$250 in credits for Cloud Sandboxes. Claim credits →

hub Explore My Hub

Search Docker Hub

CtrlK

imjk04 Your individual account

Repositories / pr2 / General

imjk04/pr2

Last pushed 1 minute ago · Repository size: 262.1 MB · ☆0 · ↓0

Add a description

Add a category

General Tags Image Management Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
app	linux	Image	less than 1 day	1 minute
db	linux	Image	less than 1 day	1 minute

DOCKER SCOUT INACTIVE
Activate

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team's builds.

Upgrade subscription

Gmail

YouTube

Update User Profile

Name :

jayakrishna

Email :

jayakrishna@gmail.com

Date Of Birth :

eg.01/01/1998

Father's Name :

Father's Name

Mother's Name :

Mother's Name

Gender

☒ Male
☐ Female
☐ Other

Marital Status:

☐ Married
☒ Unmarried

Permanent Address :

Area,Street,Dist

Temporary Address :

Area,Street,Dist

Primary Occupation :

Primary Occupation

Secondary Occupation :

Secondary Occupation

Skills :

Skills

Phone Number :

Phone Number

```

root@manager:~# docker stack ls
NAME                SERVICES
javaapp             2
[root@manager ~]# docker stack services javaapp
ID                  NAME                MODE                REPLICAS    IMAGE                PORTS
z5zravg7ch2r       javaapp_application  replicated          3/3          imjk04/pr2:app      *:1111->8080/tcp
upiq4sghjelv       javaapp_devopsdb    replicated          1/1          imjk04/pr2:db       *:3306->3306/tcp
[root@manager ~]# docker service ps javaapp_app
no such service: javaapp_app
[root@manager ~]# docker service ps javaapp_application
ID                  NAME                IMAGE                NODE          DESIRED STATE    CURRENT STATE
E
qr0lemfwf46x       javaapp_application.1  imjk04/pr2:app      node-1        Running           Running 10 m
inutes ago
msfrzj94hobh       javaapp_application.2  imjk04/pr2:app      node-2        Running           Running 10 m
inutes ago
qvsjdrqlo73v       javaapp_application.3  imjk04/pr2:app      manager       Running           Running 10 m
inutes ago
[root@manager ~]#

```

```

root@manager:~#
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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases; use accounts;
+-----+
| Database |
+-----+
| information_schema |
| accounts |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables; select * from user;

```

```
root@manager:~  
L      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
NULLL  | NULL      | NULL      | NULL      | $2a$11$6BSmYPrT8I8b9yHmx.uTRu/QxnQM2vhZYQa8mR33a  
ReWA4WFiHyGK |  
| 14 | imjk04      | imjk04@gmail.com      | NULL      | NULL      | NULL      | NULL  
L      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
NULLL  | NULL      | NULL      | NULL      | $2a$11$rw54b3Ib8EfuZRxbJYiKE002eC4/ukseI21Bduzxl  
Q3kTQf71upuq |  
| 15 | jayk123      | jayk123@gmail.com      | NULL      | NULL      | NULL      | NULL  
L      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
NULLL  | NULL      | NULL      | NULL      | $2a$11$paRtKKpm6sc5ml4ey.fc3en7wVy3zr8wh5a.ZiI6D  
bhPPD4so67eG |  
| 16 | jayakrishna | jayakrishna@gmail.com | NULL      | NULL      | NULL      | NULL  
L      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL      | NULL  
NULLL  | NULL      | NULL      | NULL      | $2a$11$/94OS2xklBvFA6YixFh3S.BagjVfD5Ix52xFckSV2  
Tht.5oIWUerG |  
+-----+-----+-----+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+-----+-----+-----+-----+  
+-----+  
10 rows in set (0.00 sec)  
  
mysql> █
```